

4-1.

直接计算: N^2 次乘法, $N(N-1)$ 次相加

FFT 计算: 一次蝶形 $\begin{cases} 1 \text{ 次乘法} \\ 2 \text{ 次相加} \end{cases}$ 共有 $\frac{N}{2} \log_2 N$ 次蝶形

时间 $T_D = 4N^2 + N(N-1) = 5.241856 \text{ ms}$

$$T_F = 4 \times \log_2 N \times \frac{N}{2} + 2 \times \frac{1}{2} \times \log_2 N \times \frac{N}{2} = 30.72 \text{ ms}$$

快速卷积时间: $2T_F + 1024 \times 4 \mu\text{s} = 65.536 \text{ ms}$

$$\lim f_s = \frac{1024}{65.536 \times 10^{-3}} = 15625 \text{ Hz}$$

$$\lim f_{c_{\text{max}}} = \frac{f_s}{2} = 7.8125 \text{ kHz}$$

4-2

直接计算: $T_D = 10N^2 + 10N(N-1) = 10.96128 \text{ ms}$

FFT: $T_F = 10 \times \log_2 N \times \frac{N}{2} + 2 \times 10 \times \log_2 N \times \frac{N}{2} = 0.1536 \text{ ms}$

快速卷积时间: $2T_F + 1024 \times 10.15 = 0.31744 \text{ ms}$

$$\lim f_s = \frac{1024}{0.31744 \times 10^{-3}} = 3.23 \text{ MHz}$$

$$\lim f_c = \frac{f_s}{2} = 1.61 \text{ MHz}$$

4.4

$$\underline{x}_1(t) = x(t) \quad , \quad t = 0, \dots, N-1$$

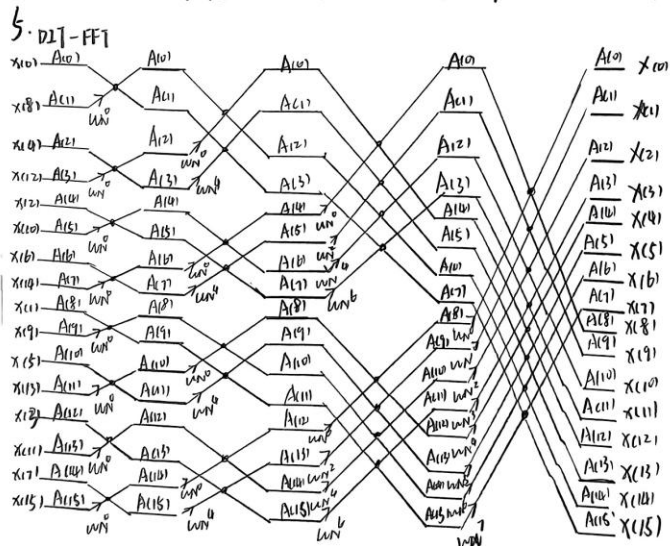
$$x_2(t) = x(t+1) \quad , \quad t = 0, \dots, N-1$$

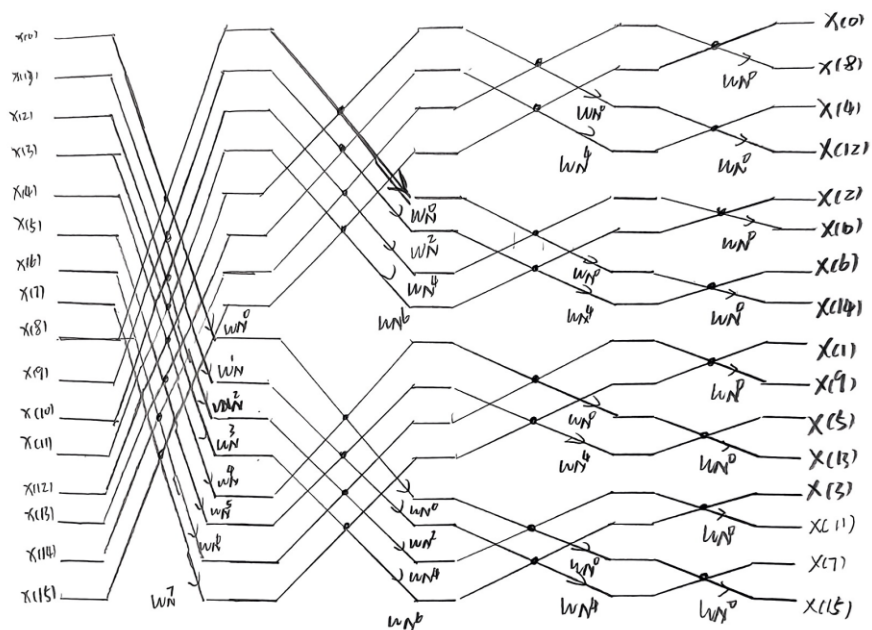
$$\begin{aligned} \text{则} \quad \sum_{n=0}^{2N-1} x(n) W_N^{kn} &= \sum_{n=0}^{N-1} x(n) W_N^{kn} + \sum_{n=0}^{N-1} x(n) W_N^{k(n+N)} \\ &= \sum_{r=0}^{N-1} x_1(r) W_N^{kr} + \sum_{r=0}^{N-1} x_2(r) W_N^{k(2r+1)} \\ &= \sum_{r=0}^{N-1} x_1(r) W_N^{kr} + W_N^k \sum_{r=0}^{N-1} x_2(r) W_N^{kr} \\ &= x_1(k) + W_N^k x_2(k) \quad , \quad k = 0, \dots, N-1 \end{aligned}$$

$$x_1(0) = x_1(N) \quad , \quad x_2(N) = x_2(0) \quad W_N^N = -1$$

∴ 由上式可计算出 $N+1$ 点。

利用实序列 DFT 具有共轭对称性，则 $x_{(N-k)} = x^*(k) \quad , \quad k = 1, \dots, N-1$





共 $N=16$ 点, $\log_2 16 = 4$ 层

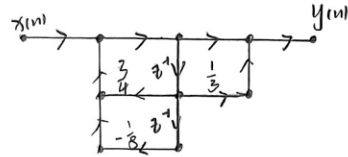
则共有 $\frac{N}{2} \log_2 N = 32$ 次复乘.

DI-FFT - DIF-FFT 次数一致

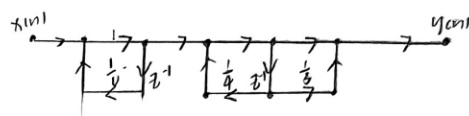
6-1.

$$H(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}} = \frac{z(z + \frac{1}{2})}{(z - \frac{1}{2})(z - \frac{1}{4})} = \frac{\frac{10}{3}z}{z - \frac{1}{2}} + \frac{-\frac{7}{3}z}{z - \frac{1}{4}}$$

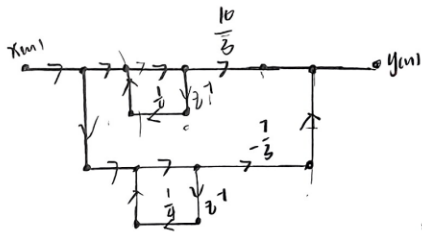
直接型:



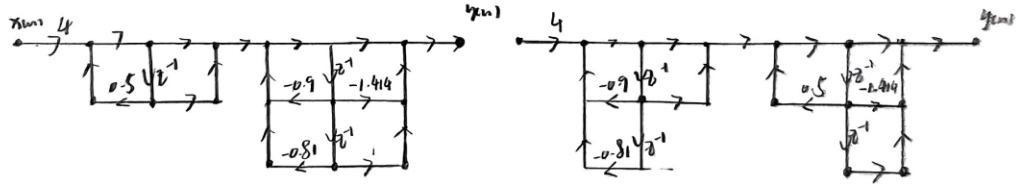
级联型



并联型



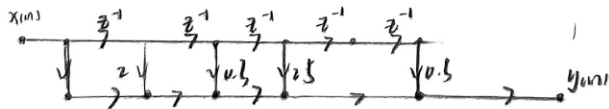
6-4



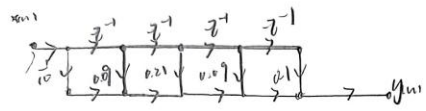
由于公式一采用的延时器较少

6-8

$$H(z) = 1 + 2z^{-1} + 0.5z^{-2} + 2.5z^{-3} + 0.5z^{-5}$$



6-9.



6-10

